

Unit 7, Lesson 15: Evaluating Exponents

Benchmark

MA.6.NSO.3.3 Evaluate positive rational numbers and integers with natural number exponents.

Learning Targets

- ☐ I can evaluate integers with natural number exponents.
- ☐ I can evaluate a positive rational number raised to a power.

7.15.1 Warm-Up: Consider the YET

1. Consider the belief that “you can get better at whatever you want to get better at.” Explain whether you agree or disagree.
2. Consider the quote, “I’ve never heard of someone who took up running and didn’t get faster the more they did it.” That person just may not have been a fast runner ... YET.

Complete the statement to describe something you want to get better at.

I _____

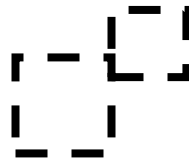
_____...YET!

7.15.2 Exploration: Building on the Understanding of Exponents

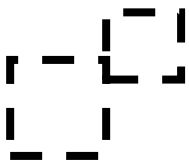
Recall the vocabulary of exponential expressions.

exponent (or power)
 base
 value (of the expression)
 $3^4 = \underbrace{3 \cdot 3 \cdot 3 \cdot 3}_{\text{factors}} = 81$

1. Think of a number that will result in a value of 16 when it is raised to a power less than 6.
2. Write the base and its exponent.



3. Compare your expression with your partner and check for accuracy.
4. If your partner found a different base and exponent that meets the description, write it below. If your partner wrote the same base and exponent as you, work together to find a second exponential expression that also works.



5. With your partner, choose one of the exponential expressions used and rewrite it, making the base negative.

$$(-\underline{\quad})^{2^3}$$

6. Explain whether the negative base changes the value of the expression.
7. Discuss with your partner how you could change the exponent to result in a negative answer. Write a brief summary of your discussion below.

7.15.3 Bring It Together: Building on the Understanding of Exponents

When evaluating exponents with a negative base, the same rules of multiplication with integers apply.

1. In the expression $(-4)^2$, the base is _____ and the exponent is _____, resulting in the repeated multiplication sentence of _____.
2. Why do the results of $(-4)^2$ result in a positive number?
3. In the expression $(-4)^3$, the base is _____ and the exponent is _____, resulting in the repeated multiplication sentence of _____.
4. Why do the results of $(-4)^3$ result in a negative number?
5. For each expression, determine whether its value is positive or negative.

Expression	Positive or Negative Value?
$(-2)^2$	
$(-2)^3$	
$(-2)^4$	
$(-2)^5$	
$(-2)^{19}$	

6. When evaluating exponents with a negative base, the value of the expression will be

☐ positive

☐ negative

 if the exponent is an

☐ odd

☐ even

 number, and the value of the expression will be

☐ positive

☐ negative

 if the exponent is an

☐ odd

☐ even

 number.

7.15.4 Your Turn!

1. Evaluate each expression.

a. $(-3)^4$

b. $\left(\frac{15}{17}\right)^2$

c. $(-1)^5$

2. Rewrite each repeated multiplication sentence as an exponential expression. Then, find the value of the expression.

a. $(-12)(-12)(-12)$

b. 3.4×3.4

3. Determine if each equation in the table is true or false. Show your work to support your answer.

Equation	True or False?	Work
$16^1 = (-8)^2$		
$(-5)^3 = -5 \cdot 5 \cdot 5$		

7.15.5 Wrap-Up: Reflection

1. In this unit, you have learned multiple concepts about operations with integers. Rate yourself on the following skills based on your level of understanding.

Statement	Rating 1 – I struggle with this. 2 – I have some skills. 3 – I am almost there. 4 – I have got this!
I can evaluate positive rational numbers and integers raised to a power.	
I can express composite whole numbers as a product of prime factors.	
I can add integers.	
I can subtract integers.	
I can multiply integers.	
I can divide integers.	

